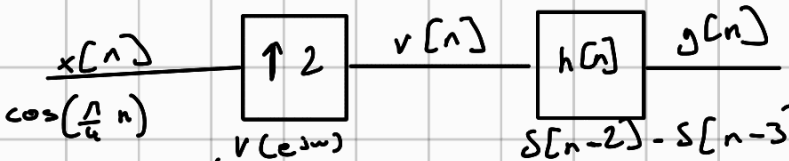
Quiz Çözümü



$V(e^{jw})$

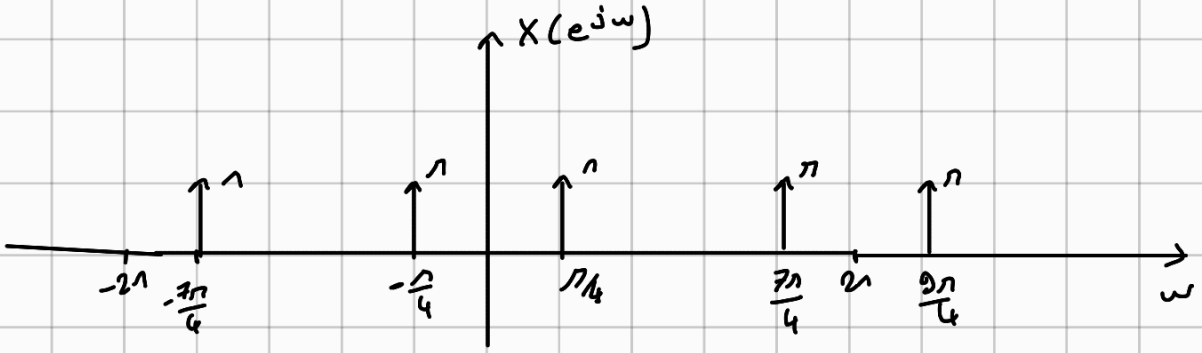
$v[n] = ?$

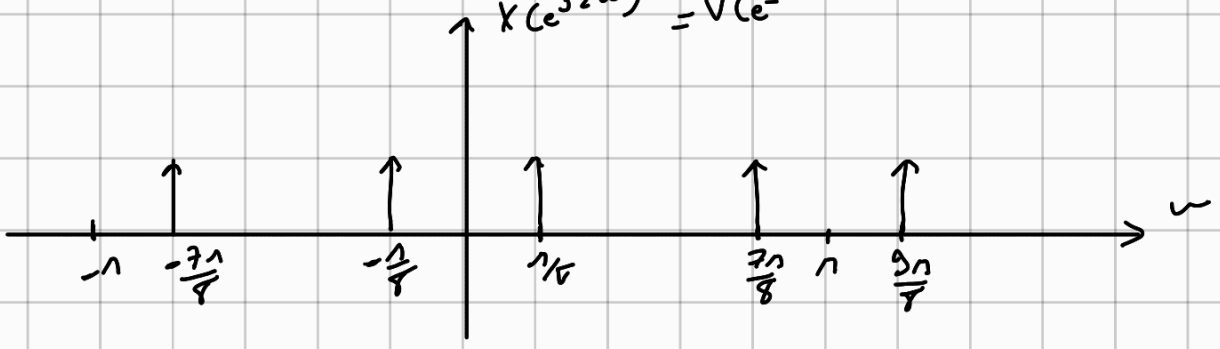
$s[n-2] - s[n-3]$

$H(e^{jw}) = e^{-j2w} - e^{-j3w}$

$V(e^{jw}) = X(e^{j2w})$

$V(z) = X(z^2)$





$v[n] = \cos\left(\frac{\pi}{8} n\right) + \cos\left(\frac{7\pi}{8} n\right)$

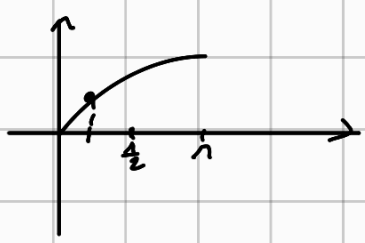
$H(e^{jw}) = e^{-j2w} - e^{-j3w} = 2j\, e^{-j\frac{5}{2}w} \left(\frac{e^{j\frac{1}{2}w} - e^{-j\frac{1}{2}w}}{2j}\right)$

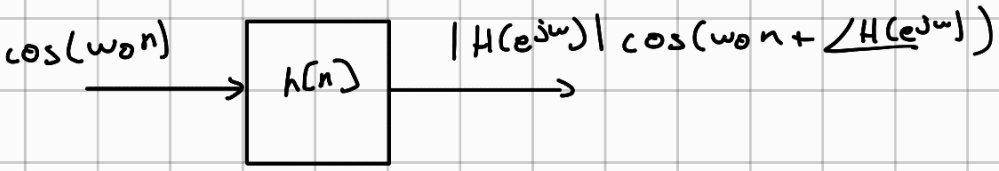
$= e^{j\frac{\pi}{2}} e^{-j\frac{5}{2}w} \; 2\sin\left(\frac{w}{2}\right)$

$e^{-(\frac{5}{2}w - \frac{\pi}{2})}$

$|H(e^{jw})| = 2\left|\sin\frac{w}{2}\right|$

$\angle H(e^{jw}) = -\frac{5}{2}w + \frac{\pi}{2} \qquad 0 < w < \pi$





FIR Filter Design Using windows

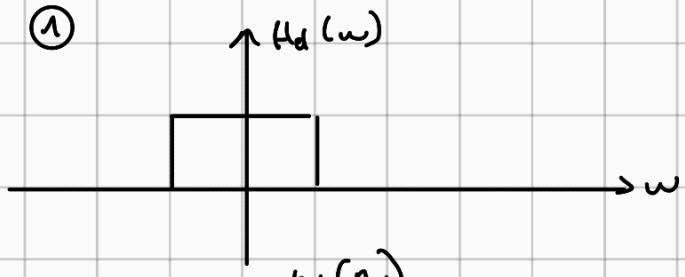
① start with desired freq response $H_d(\omega)$

② compute $IDTFT\{H_d(\omega)\} = h_d[n]$

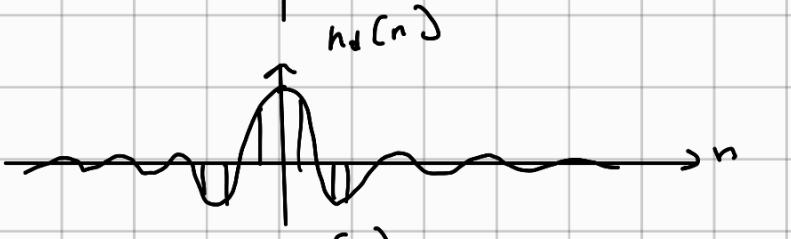
③ $h[n] = h_d[n]\,w[n]$

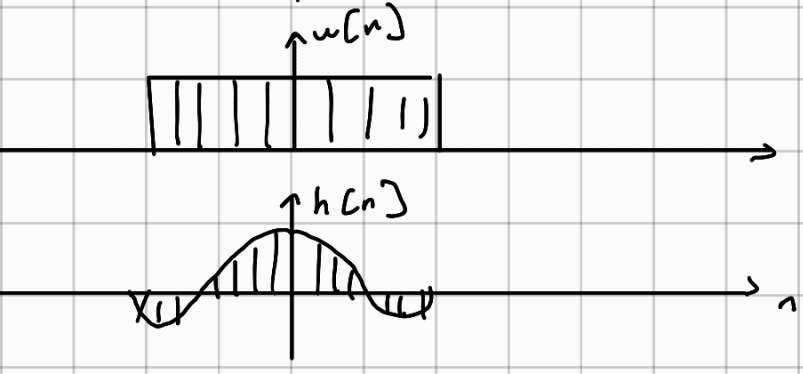
$H(e^{j\omega})$, $H(\omega)$ } Ayrık

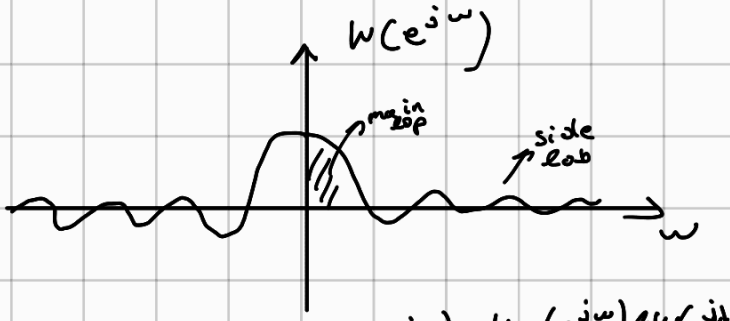
$H(-\Omega)$, $H(j\Omega)$ } sürekli

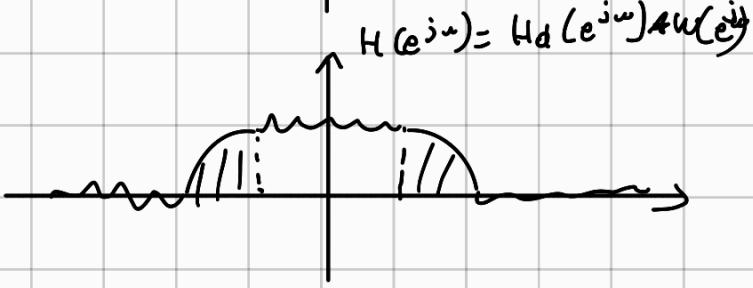


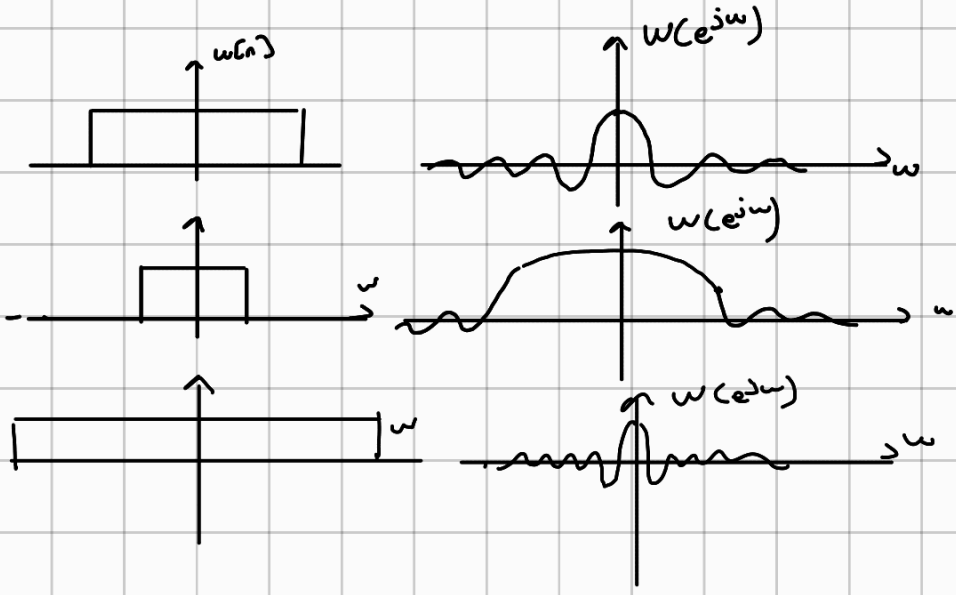


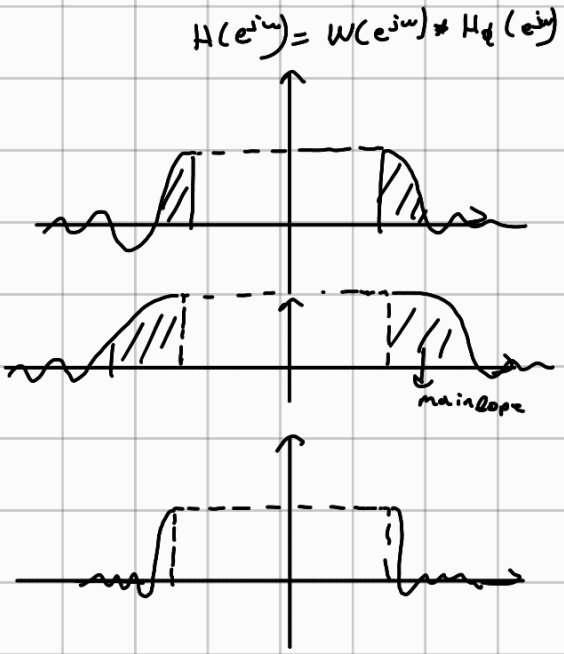












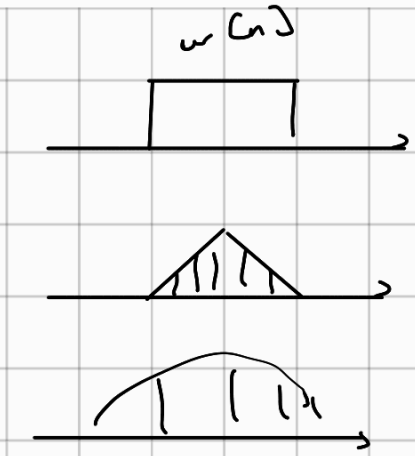




Fig 4.14

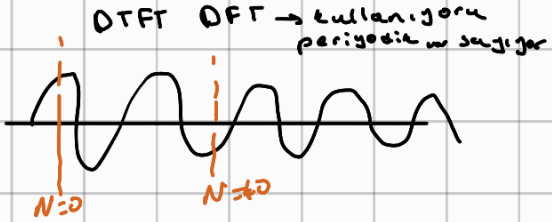
Üçgen kullanıldığında side loblar daha az

main loob ne kadar dar olursa daha iyi;

1- shape → main lobe
→ side lobe ①

2- size → main lobe ②

DTFT DFT → kullanıyoruz periyodik mi sayılır





resolüsyon bileşenlerin birbirine karışımı

IIR Filter Design

Transformation based
- Impulse invariance
- bilinear transformation

1. Given set of digital filter spes
2. **Transform** specs to analog filter-specs
3. Design analog filter $H_a(s)$
4. **Transform** $H(s) \rightarrow H(z)$

Impulse Invariance

$h_a(t)$

$h_d[n]$

$h_d[n] = h_a(t)\big|_{t=nT}$

$H(z) = Z[h_d[n]]$

Ex:

$H_a(s) = \frac{A}{s+a}$

$h_a(t) = Ae^{-at}u(t)$

$h_a[n] = Ae^{-aTn}$

Ex:

$H_a(s) = \frac{A}{s+a}$

$h_a(t) = Ae^{-at}u(t)$

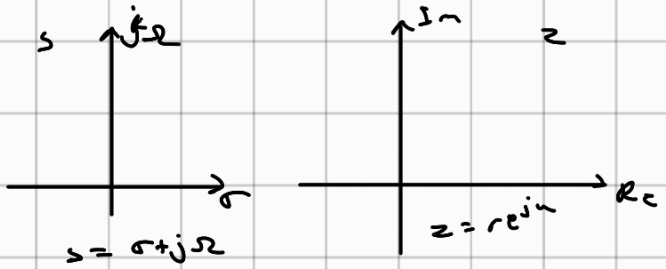
$h_d[n] = Ae^{-aTn}u(nT)$

$= A(e^{-aT})^n u[n] \Rightarrow H(z) = \frac{A}{1-e^{-aT}z^{-1}}$

$$H_a(s) = \frac{Y(s)}{X(s)} = \frac{\sum_{k=0}^{M} b_k s^k}{\sum_{k=0}^{N} a_k s^k}$$

$$H_{LP}(s) = \frac{1}{s^3+2s^2+2s+1} = \frac{Y(s)}{X(s)}$$

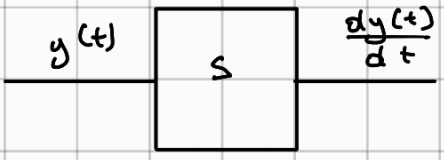
$$y'''(t) + 2y''(t) + 2y'(t) + y(t) = x(t)$$

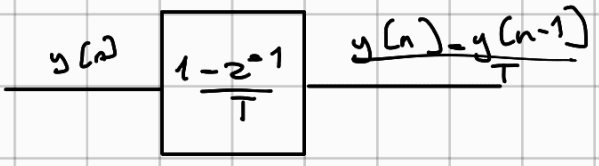


Filter Design by Approximation of Derivation

$$\left.\frac{dy(t)}{dt}\right|_{t=nT} = \frac{y(nT) - y((n-1)T)}{T} = \frac{y[n] - y[n-1]}{T}$$

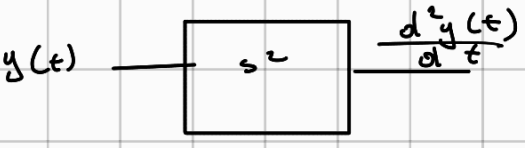


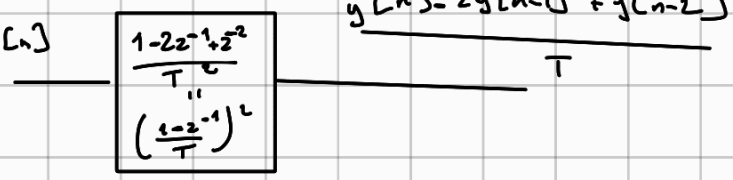


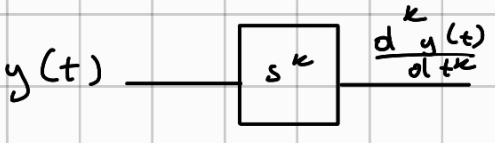


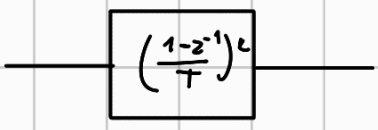
$$\left.\frac{d^2y(t)}{dt^2}\right|_{t=nT} = \frac{d}{dt}\left(\left.\frac{dy(t)}{dt}\right)\right|_{t=nT} = \frac{y(nT) - 2y((n-1)T) + y((n-2)T)}{T^2}$$

$$\frac{y[n] - 2y[n-1] + y[n-2]}{T^2}$$





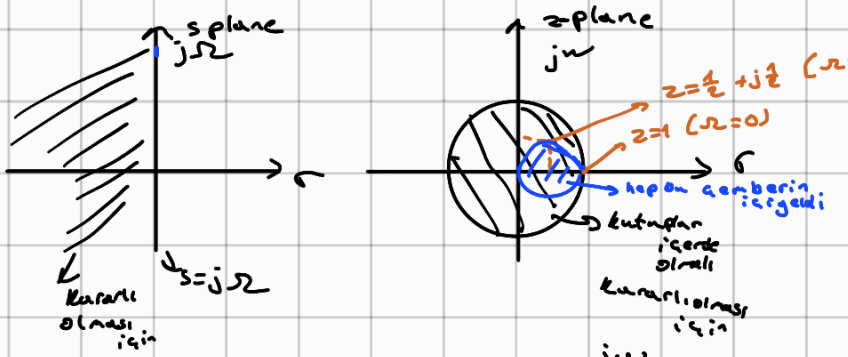




$$s = \frac{1 - z^{-1}}{T}$$

$$H(z) = H_a(s)\Big|_{s \leftarrow \frac{1-z^{-1}}{T}}$$

$$s = \frac{1-z^{-1}}{T} \Rightarrow z = \frac{1}{1-sT}$$



$$z = \underbrace{\sigma}_{<1} e^{jw}$$

$s = \sigma + j\Omega$

$s = j\Omega$ sınır

$$z = \frac{1}{1-sT}\Big|_{s=j\Omega} = \frac{1}{1-j\Omega T} \cdot \frac{1+j\Omega T}{1+j\Omega T}$$

Burdasınırda birkutup orda (z'de) nerde

$$= \frac{1}{1+\Omega^2T^2} + j\frac{\Omega T}{1+\Omega^2T^2}$$

$\Omega = 0 \Rightarrow z = 1$

$\Omega = \frac{1}{T} \Rightarrow z = \frac{1}{2} + \frac{1}{2}j$

$\Omega = \infty \Rightarrow \frac{1}{T}$

Çekiği bir yöntem değilmiş.

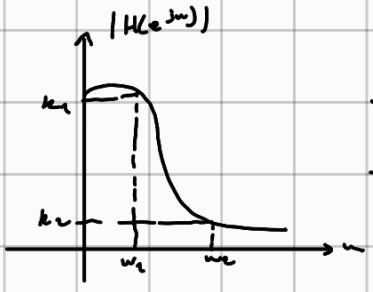
IIR Filter Design by the Bilinear Transformation

$$s = \frac{2}{T}\frac{1-z^{-1}}{1+z^{-1}} \Rightarrow z = \frac{1-s\frac{T}{2}}{1+s\frac{T}{2}}$$

$$H_a(s)\Big|_{s \leftarrow \frac{2}{T}\frac{1-z^{-1}}{1+z^{-1}}} = H(z)$$

$$\Omega = \frac{2}{T}\tan\left(\frac{w}{2}\right)$$

$$w = 2\tan^{-1}\left(\frac{\Omega T}{2}\right)$$



$$\Omega_1 = \frac{2}{T}\tan\left(\frac{w_1}{2}\right)k_1$$

$$\Omega_2 = \frac{2}{T}\tan\left(\frac{w_2}{2}\right)k_2$$

$$H_a(s)\Big|_{s \leftarrow \frac{2}{T}\frac{1-z^{-1}}{1+z^{-1}}}$$

